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Nitte (Deemed to be University)

(Established under Sec (3) of UGC ACT,1956)

First Semester MCA (CBCS) Degree Examinations, Dec. 2025 - Jan. 2026

25MCA102 - ADVANCED DATABASE SYSTEMS

Duration: 3 Hours]

[Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from **each Unit**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Describe Entity types, Attributes, and Keys with examples.	10	L2	1	1,3
b)	Design an ER diagram for a university database that includes students, courses, instructors, and prerequisites. Indicate weak entities and participation constraints.	10	L3	1	2,3
2. a)	Define DBMS. Explain the advantages and disadvantages of DBMS approach.	10	L2	1	1,3
b)	Evaluate the importance of relationship constraints such as cardinality and participation in maintaining data integrity in ER models.	10	L5	1	2,3
Unit – II					
3. a)	Illustrate the use of Outer Joins with examples and queries.	10	L3	2	1,3
b)	Demonstrate with an example how referential integrity constraints and cascading options (SET NULL, CASCADE, SET DEFAULT) preserve consistency during updates and deletions.	10	L4	2	2
4. a)	Apply SQL aggregate functions and demonstrate their use through suitable example queries.	10	L3	2	1,3
b)	Write a complete schema definition using DDL statements (CREATE, ALTER, DROP) and justify each constraint applied (NOT NULL, CHECK, UNIQUE, etc.).	10	L4	2	2
Unit – III					
5. a)	Demonstrate the role of COMMIT, ROLLBACK, and SAVEPOINT in maintaining transactional consistency with a suitable real-time example.	10	L3	3	2,3
b)	Write the syntax of stored procedure in My SQL. Create a stored procedure to calculate total salary of employees in a database.	10	L2	3	4
6. a)	Explain the use of cursors in MySQL procedural programming. Illustrate a code example that uses a cursor to iterate through a result set and perform updates conditionally.	10	L3	3	2,3
b)	Write a stored function and a trigger for an employee database to automatically update bonus details whenever a new salary record is inserted.	10	L2	3	4
Unit – IV					
7. a)	Explain 1NF, 2NF and 3NF with examples.	10	L2	4	1,3
b)	Analyze the properties of transactions (ACID) and explain how they ensure reliable transaction processing in concurrent systems.	10	L4	4	2,5



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| 8. a) | Explain the informal design guidelines for relational schema design. How do they help in preventing anomalies and ensuring database consistency? | 10 | L2 | 4 | 1,3 |
| b) | Analyze transaction states with neat diagram. | 10 | L4* | 4 | 2 |

Unit – V

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|--------|---|----|----|---|-----|
| 9. a) | What is MongoDB? Explain features with examples. | 10 | L2 | 5 | 1,3 |
| b) | Analyze JSON support and unique key generation in MongoDB with suitable examples. | 10 | L4 | 5 | 2 |
| 10. a) | Explain advantages of NoSQL over SQL. | 10 | L2 | 5 | 1,3 |
| b) | Design and write MongoDB queries to insert, update, delete, and retrieve documents. | 10 | L4 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.

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1,3

2,3

1,3

2

1,3

2

2,3

4

2,3

4

1,3

2,3